Temperature-driven reversible *n*- to *p*-type conductivity switch in Bi-doped SnSe

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In this work, we study the temperature-dependent Seebeck coefficient and electrical conductivity of pristine and Bi-doped SnSe to clarify the microscopic origin of carrier-type reversal in lightly doped compositions. Pristine SnSe remains *p*-type over the full temperature range, while Bi substitution drives *n*-type transport. The 2% Bi-doped sample exhibits negative Seebeck coefficients at low temperature, followed by an *n*-to-*p* polarity reversal near 550 K, consistent with bipolar transport and carrier compensation, whereas the 6% Bi-doped sample remains robustly *n*-type up to 880 K. The experimental data are analyzed using a self-consistent two-carrier transport model with a smooth interpolation across the *Pnma*-to-*Cmcm* structural transition. To connect the extracted transport trends to electronic structure evolution, we perform first-principles calculations for Bi-doped SnSe in both phases. The calculated band structures show that polarity switching at low Bi concentration arises from the temperature-driven evolution of the band edges and the Fermi level, along with a progressive narrowing of the band gap that promotes intrinsic excitation. Electron-phonon self-energy calculations further indicate a monotonic reduction of the *Pnma* band gap with increasing temperature, providing a direct electronic mechanism for the onset of bipolar transport prior to completion of the structural transition.

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I. INTRODUCTION

The ability to dynamically control the dominant charge-carrier type in a semiconductor offers an alternative to conventional static doping strategies. In classical semiconductor technologies, *p*-type and *n*-type regions are permanently defined through chemical substitution, diffusion, or implantation, which fixes device functionality at the fabrication stage and imposes rigid design constraints. In contrast, materials capable of reversibly switching between hole- and electron-dominated transport in response to an external stimulus—such as temperature, electric field, or structural reconfiguration—offer a fundamentally different degree of freedom [1,2]. Such switchable semiconductors enable adaptive control over electronic polarity within a single compound, eliminating the need for spatially separated doped regions. This concept opens pathways toward simplified device architectures, reconfigurable electronics, and position-independent junctions, while also creating opportunities in areas where selective access to electrons or holes is critical, including thermoelectric energy conversion, catalysis, sensing, and multifunctional electronic components [1]. From a broader perspective, carrier-type switching reflects an interplay between electronic structure, defect chemistry, and lattice dynamics. Therefore, it provides a powerful platform for exploring coupled structural-electronic phenomena in solids and for designing materials with tunable, stimulus-responsive transport properties.

Reversible carrier-polarity switching has been experimentally observed in only a limited number of semiconductor systems, typically under conditions where the electronic structure is strongly perturbed by an external stimulus. Early demonstrations relied on temperature-driven structural transformations or highly mobile sublattices, as exemplified by $Ag_{10}Te_4Br_3$ [3] and $AgBiSe_2$ [4], in which thermally induced lattice reorganization enables reversible inversion of the dominant carrier type without changing the overall composition. These systems established that carrier polarity can be actively modulated when subtle changes in atomic arrangement or site occupancy reshape the electronic structure.

More recent studies indicate that polarity switching can also emerge in the absence of a distinct structural phase transition, driven instead by competing carrier populations whose relative contributions evolve with temperature, excitation, or microstructure. In this context, SnSe has attracted particular attention because polarity switching has been reported under remarkably diverse conditions, including some that appear counterintuitive [5–10]. Notably, Seebeck coefficient sign reversals ($p \rightarrow n$) have been observed in undoped polycrystalline SnSe, despite its intrinsically *p*-type nature [7]. Conversely, *n*-type SnSe obtained via aliovalent doping has been shown to revert to *p*-type conduction at elevated temperature [10]. Additional complexity arises from strong dependence on material form and stimulus: Bi-doped single-crystalline SnSe exhibits stable *n*-type behavior [11], whereas its polycrystalline counterpart displays temperature-driven *n*-to *p*-type conductivity transitions whose onset varies with dopant concentration [10]. Bi-doped SnSe thin films can even exhibit wavelength-dependent polarity switching under optical excitation [8]. Together, these observations indicate that multiple, possibly competing mechanisms can drive polarity switching in SnSe, rendering the underlying physics ambiguous. Therefore, understanding the mechanism underlying this

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change in polarity is crucial to the development of SnSe-based functional devices that exploit this behavior.

In this work, we systematically investigate carrier-polarity switching in bulk polycrystalline $Sn_{1-x}Bi_xSe$ ($x = 0.00$, 0.02, and 0.06) by combining temperature-dependent thermoelectric measurements with first-principles density functional theory (DFT) calculations. By tracking the evolution of the Seebeck coefficient across composition and temperature, we demonstrate that the *n*-to-*p* polarity reversal observed in lightly Bi-doped SnSe is not governed solely by abrupt structural transformations or extrinsic microstructural effects. Instead, our results reveal that switching behavior is primarily controlled by the intrinsic temperature dependence of the electronic structure, in which band-gap evolution and associated Fermi-level repositioning progressively activate minority carriers at elevated temperatures.

II. EXPERIMENTAL DETAILS

For bulk SnSe samples, Sn and Se powders ($\sim$100 mesh, $\geqslant$99.5 Sigma Aldrich) were combined in a 1:1 molar ratio, resulting in a total mass of 5 g. For Bi-doped SnSe samples, the Sn and Bi (polycrystalline lump, 99.9%, Fisher Scientific) powders were stoichiometrically balanced to incorporate 2% and 6% Bi, after which the mixture was combined with Se powders in a 1:1 ratio, also totaling 5 g. In the remainder of the paper, the Bi-doped SnSe samples are called $Sn_{1-x}Bi_xSe$ (where $x = 0.00$, 0.02, and 0.06). The prepared powder mixture was first loaded into quartz ampules, which were then evacuated to 10^{-5} mbar and sealed. These sealed ampules were placed inside a 1-in.-thick quartz tube positioned on a horizontal furnace, and the system was pumped down to a base pressure of 10^{-9} mbar to ensure sealing if the inner ampule broke. The furnace was heated slowly for over 8 h up to 1200 K, held at this temperature overnight, and then naturally cooled down to room temperature.

The synthesized samples were ground into a fine powder using a mortar and pestle and divided into portions for chemical verification and thermoelectric characterization. The structural and chemical analyses were carried out using scanning transmission electron microscopy (S/TEM), x-ray diffraction (XRD), and x-ray photoelectron spectroscopy (XPS). Approximately 2.5 g of each powdered sample were then shaped into a disk for subsequent thermoelectric characterization. To achieve this, we employed a half-inch stainless steel die set and applied a pressure of $\sim 7 \times 10^8$ Pa overnight. The resulting disk samples measured between 12.7 and 12.8 mm in diameter and 3.0–4.0 mm in height. Experimental density calculations showed 95%–96% of theoretical values.

Samples' crystallinity and elemental spectral maps were obtained using S/TEM. Measurements were conducted on a ThermoFisher TALOS F200X G2 equipped with an X-CFEG and SuperX-G2 x-ray detectors. The S/TEM was operated at 200 kV for all experiments. Sample preparation involved suspending $Sn_{1-x}Bi_xSe$ powders in ethanol, sonicating for 10 min, and drop-casting onto 3-mm Holey Carbon Film TEM grids (Electron Microscopy Services) with a carbon support layer less than 3 nm thick. After drop-casting, the samples were dried under vacuum for at least 24 h. Once dried, the prepared grids were placed on a low-background double-tilt holder stage and inserted into the TEM. Before scanning, the samples underwent a beam shower for approximately 40–60 min to stabilize any organic ligands that may affect the measurement.

The crystal structure and temperature-dependent phase changes of $Sn_{1-x}Bi_xSe$ ($x = 0.00$ and 0.02) samples were analyzed utilizing a Rigaku SmartLab diffractometer with a Mo Kα x-ray source ($\lambda = 0.7093$ Å) in a controlled atmosphere within a rough-pumped, isolated chamber. Data was collected in 100-K increments from room temperature to 873 K, with the x-ray tube operating at 50 kV and 40 mA. Room-temperature XRD measurements were performed using a Rigaku SmartLab 3-kW diffractometer with a Cu Kα x-ray source ($\lambda = 1.5406$ Å), operating at 40 kV and 44 mA. Rietveld refinement was used to analyze the diffraction patterns quantitatively, providing detailed information on lattice parameters and phases.

XPS analysis was conducted using a PHI-5400 XPS system, operated under ultrahigh vacuum conditions with a base pressure of 2×10^{-10} mbar. The study employed a nonmonochromatized Mg Kα x-ray source, producing photons at 1253.6 eV. Survey scans were performed at a pass energy of 89.45 eV, with a step size of 0.5 eV, consisting of 20 sweeps with a dwell time of 50 ms per step. A pass energy of 17.9 eV was used for high-resolution scans, featuring a finer step size of 0.025 eV and 20 sweeps with a dwell time of 50 ms per step. The peak fitting was performed using the Voigt function and Shirley-type background subtraction in the KHERVEFITTING software [12]. Electrical conductivity and the Seebeck coefficient were measured simultaneously using a four-probe method with a Linseis LSR-3 system in a helium atmosphere at temperatures ranging from 300 to 900 K. Before measurements, the chamber was purged three times with helium to minimize oxygen contamination, then filled with helium at a pressure of 1.1 bar. The disk-shaped samples were placed vertically between the upper and lower molybdenum electrodes, aligned along their diameters. Type-S thermocouple probes, capped with gold foil, were positioned in contact with the samples, maintaining a probe distance of 7–8 mm, which was accurately measured using a camera and automated software. A 50-K temperature gradient was applied between the electrodes, resulting in 7–11-K differences between the upper and lower probes during the measurement. A constant direct current of 1 mA was applied to the sample to assess electrical conductivity, assuming an almost ideal one-dimensional current flow.

III. COMPUTATIONAL DETAILS

First-principles calculations were performed using the vienna ab initio simulation package (VASP) within the projector-augmented-wave (PAW) framework [13,14] and the Perdew-Burke-Ernzerhof (PBE) generalized-gradient approximation [15]. Both the low-temperature *Pnma* and high-temperature *Cmcm* phases of SnSe were fully relaxed. Bi doping was modeled by substituting one or two Sn atoms in a 72-atom supercell, corresponding to Bi concentrations of 2.78% and 5.56%. Temperature-dependent band-gap renormalization for the *Pnma* phase was obtained from first-principles electron-phonon self-energy calculations. Additional computational

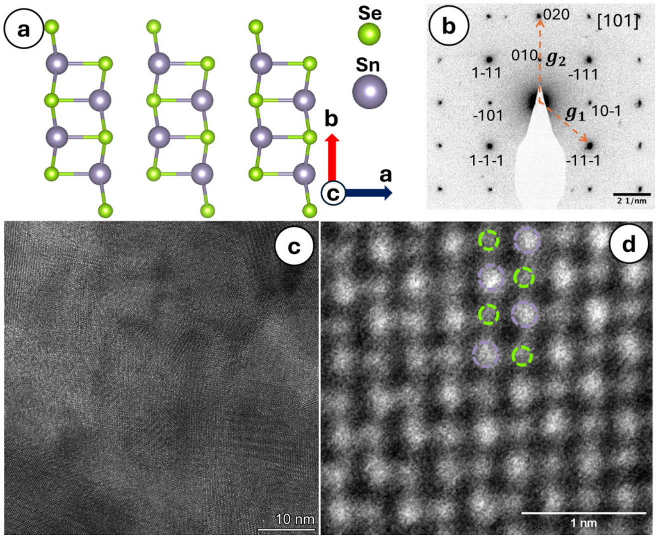

FIG. 1. Microstructural characterization of orthorhombic SnSe. (a) The crystal structure of orthorhombic SnSe is viewed through the *c* axis, illustrating its layered arrangement of Sn (larger gray spheres) and Se (smaller green spheres) atoms. (b) SAED pattern along the [101] zone axis. (c) High-resolution TEM (HR-TEM) image of the undoped SnSe sample, showing different crystalline domains. (d) HAADF-STEM image acquired close to the [001] zone axis. The crystallographic a and b axes are indicated to facilitate direct comparison with the structural model in (a). Sn atomic columns appear brighter due to Z contrast, while Se columns appear weaker.

details and transport modeling are provided in the Supplemental Material [16] (see also Refs. [17–21] therein).

IV. RESULTS AND DISCUSSION

Figure 1(a) shows the crystal structure of orthorhombic SnSe along the *c* axis, demonstrating the layered arrangement of Sn and Se atoms. Figure 1(b) displays the selected area electron diffraction (SAED) pattern along the [101] zone axis. The measured reciprocal distances of $|\boldsymbol{g}_1| = 3.369\ \mathrm{nm}^{-1}$ and $|\boldsymbol{g}_2| = 4.460\ \mathrm{nm}^{-1}$ agree closely with the simulated values of 3.408 and 4.499 nm^{-1}(see Fig. S1(a) in the Supplemental Material [16]). All observed reflections can be consistently indexed to the *Pnma* structure. Figure 1(c) shows a high-resolution TEM (HR-TEM) image, confirming the polycrystalline nature of the SnSe samples with visible grain boundaries. Figure 1(d) shows a high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) image acquired near the [001] zone axis, aligned with the structural projection in Fig. 1(a). The atomic columns are consistent with the orthorhombic SnSe structure [22], with brighter Sn columns and weaker Se columns due to Z-contrast imaging. The corresponding simulated image is shown in Fig. S1(b) in the Supplemental Material [16].

Figure 2 summarizes the compositional and chemical analysis of $Sn_{1-x}Bi_xSe$ samples. In Fig. 2(a), energy dispersive x-ray spectroscopy (EDS) maps of Bi, Sn, and Se elements correlate with the HAADF image of the $Sn_{1-x}Bi_xSe$ sample ($x = 0.06$), confirming the uniform distribution of all

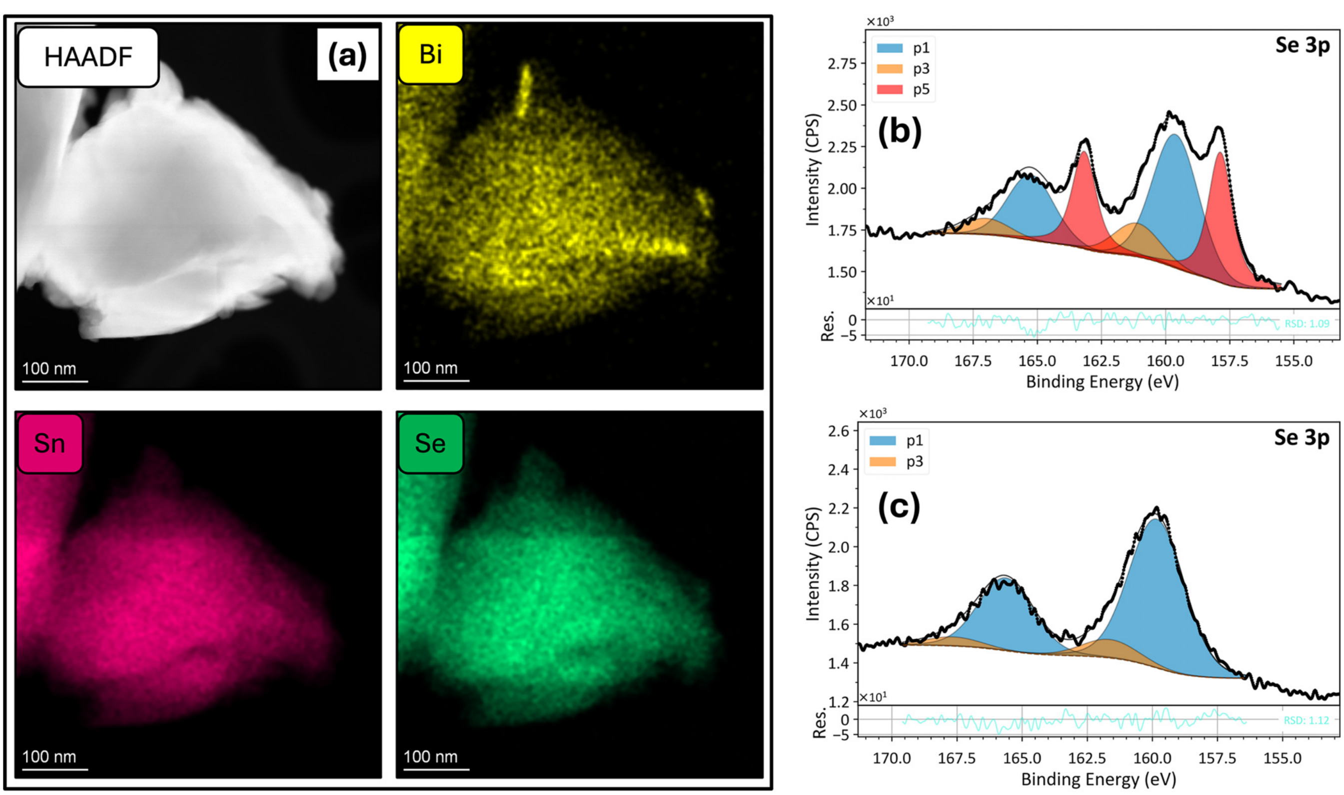

FIG. 2. Compositional and chemical analysis of Bi-doped SnSe. (a) HAADF image and corresponding EDS elemental maps for Bi, Sn, and Se in the $Sn_{1-x}Bi_xSe$ sample ($x = 0.06$). (b) High-resolution XPS spectrum of the $Sn_{1-x}Bi_xSe$ sample ($x = 0.06$), showing Se 3*p* peaks overlaid with the Bi 4*f* peaks. (c) The high-resolution XPS spectrum of the undoped SnSe sample shows only the Se 3*p* peaks.

TABLE I. The relative atomic concentrations for $Sn_{1-x}Bi_xSe$ samples.

Sample	Sn (atomic %)		Se (atomic %)		Bi (atomic %)		O (atomic %)	
	EDS	XPS	EDS	XPS	EDS	XPS	EDS	XPS
$x = 0.00$	47.30	50.67	51.35	24.42	0.00	0.00	1.35	24.91
$x = 0.02$	48.09	52.68	46.15	21.76	0.27	0.81	5.48	24.76
$x = 0.06$	46.76	50.68	46.44	27.19	1.48	2.24	5.32	19.90

elements. Table I compares the relative atomic concentrations obtained from EDS and XPS. While EDS confirms the stoichiometric Sn:Se ratio with increasing Bi content, XPS, which is more surface sensitive, reveals higher oxygen levels, likely due to surface oxidation [23].

To further investigate the local chemical environments and verify Bi incorporation, high-resolution XPS measurements were performed (Figs. 2(b) and 2(c), and Fig. S3 and Table S1 in the Supplemental Material [16]). For pristine and Bi-doped SnSe, the Sn $3d_{5/2}$ core levels appear at ∼485.2 eV, shifted to higher binding energy relative to metallic Sn (484.5 eV) [24], consistent with Sn in Sn-Se lattice [25]. Additional components at higher binding energies (486−487 eV) can be attributed to surface-oxidized Sn species (SnO and SnO_2) [26]. The Se $3p_{3/2}$ levels exhibit two components at ∼159.6−159.8 and ∼161.1−161.6 eV, corresponding to lattice-bound Se in SnSe and a minor contribution from elemental Se, respectively [27]. In Bi-doped samples, the Bi $4f_{7/2}$ peak is observed at ∼157.9 eV, significantly shifted from metallic Bi (156.0 eV), indicating that Bi is chemically bonded within the SnSe matrix rather than present as a metallic secondary phase [28]. Although the EDS maps suggest slight Bi enrichment at some regions, XPS does not reveal any metallic Bi signature. Together, EDS and XPS confirm the incorporation of Bi atoms into the SnSe lattice, with increasing Bi content correlating with doping ratios and validating the formation of stable Sn–Se bonds.

Figure 3(a) presents the room-temperature XRD patterns of $Sn_{1-x}Bi_xSe$ samples with nominal compositions $x = 0.00$, 0.02, and 0.06. All prominent diffraction peaks can be indexed to the orthorhombic SnSe phase with space group *Pnma* (No. 62), consistent with the standard reference (PDF Card No. 50–544). The refined average lattice parameters (a = 11.523 Å, b = 4.161 Å, and c = 4.439 Å) closely match reported values [29], indicating that the primary crystal structure is preserved upon Bi incorporation. Simulated XRD patterns for various substitutional Bi and vacancy configurations yield diffraction patterns that resemble those of pristine SnSe. In contrast, Bi intercalation between SnSe layers introduces additional weak reflections that are absent in experimental data, suggesting that Bi occupation of interlayer sites is unlikely or contributes only negligibly to the measured diffraction signal (see Fig. S2(c) in the Supplemental Material [16]). Despite the polycrystalline nature of the samples, all compositions exhibit a pronounced (400) reflection, indicative of preferred orientation along the [*h00*] direction.

SnSe undergoes a displacive, two-step phase transition beginning around 600 K and completing near 800 K [30,31], evolving from a highly asymmetric *Pnma* phase at room temperature to a highly symmetric *Cmcm* phase at elevated temperatures. To investigate the effect of Bi doping on structural evolution, temperature-dependent XRD measurements were performed on the $x = 0.00$ and $x = 0.02$ samples over 300–873 K. The corresponding temperature-dependent lattice parameters extracted from these measurements are shown in Figs. 3(b) and 3(c), while the full diffraction patterns are provided in Figs. S2(a) and S2(b) in the Supplemental Material [16]. With increasing temperature, gradual shifts in diffraction

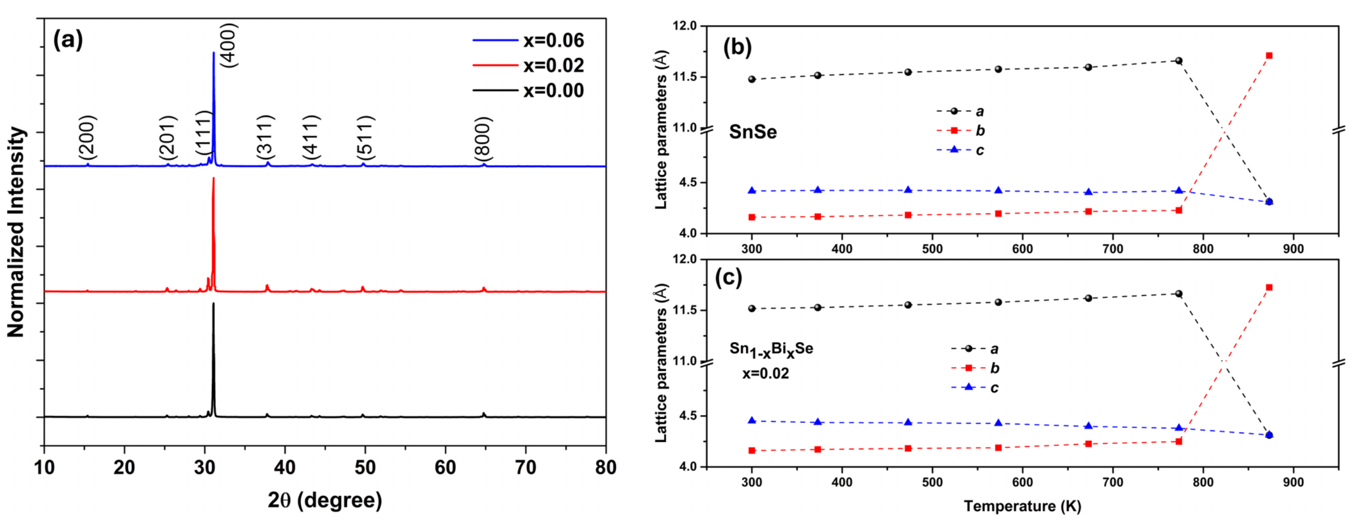

FIG. 3. Structural analysis of undoped and Bi-doped SnSe samples. (a) Room-temperature XRD patterns of $Sn_{1-x}Bi_xSe$ samples ($x = 0.00$, 0.02, and 0.06). (b) The temperature-dependent evolution of lattice parameters for the undoped sample and (c) the 2% Bi-doped SnSe sample.

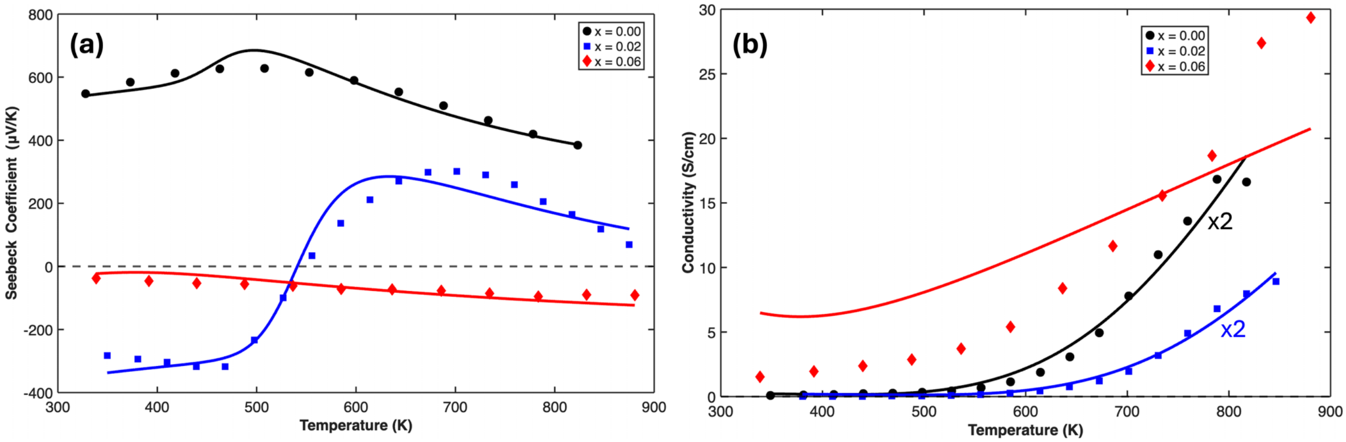

FIG. 4. Temperature-dependent transport properties. The temperature-dependent (a) Seebeck coefficient and (b) electrical conductivity for pristine SnSe ($x = 0.00$), 2% Bi-doped SnSe ($x = 0.02$), and 6% Bi-doped SnSe ($x = 0.06$). Symbols represent experimental data, and solid lines correspond to fits from the two-carrier transport model.

peak positions are observed, reflecting thermal expansion and progressive lattice distortion. The diffraction peaks exhibit slight shifts with increasing temperature, resulting in dynamic lattice parameters, as shown in Figs. 3(b) and 3(c). We observe that the lattice parameters in the b and c directions become more similar in the temperature range before the phase transition in both samples, around 820 K, consistent with the literature [32], indicating that Bi doping did not influence the phase transition in SnSe.

Figure 4 summarizes the temperature-dependent transport properties of pristine and Bi-doped SnSe samples. Pristine SnSe exhibits clear p-type behavior over the entire measured temperature range, reaching a maximum Seebeck coefficient of approximately 627.5 μV/K at 510 K. In contrast, Bi doping progressively shifts the transport toward n-type conduction. The 2% Bi-doped samples display negative Seebeck coefficients up to approximately 550 K, above which bipolar transport becomes apparent, while the 6% Bi-doped sample remains n-type over the full temperature range up to 880 K. To evaluate composition dependence, we additionally measured a 4% Bi-doped SnSe sample, which also exhibits Seebeck polarity switching (see Fig. S4(a) in the Supplemental Material [16]). In addition, heating-cooling cycle measurements verified that the observed n-to-p carrier polarity switch is reversible (see Fig. S4(b) in the Supplemental Material [16]).

To gain quantitative insight into the transport behavior of the system, we developed a transport model and used experimental data to estimate key parameters. The experimentally observed nonlinear $S(T)$ behavior in pristine and 2% Bi-doped SnSe cannot be captured within a single-carrier description, indicating the presence of multicarrier transport effects. The sign change arises from competition between electron and hole contributions, whose relative weights evolve with temperature as carrier densities and mobilities change. Because these quantities exhibit strong, often competing, temperature dependencies, identifying the transition temperature and extracting reliable parameters directly from experimental data requires a self-consistent two-carrier (electron-hole) transport framework. The effects of the *Pnma*-to-*Cmcm* phase transition on the band gap and on the electron and hole effective masses are incorporated through a smooth interpolation scheme. On the other hand, for the 6% Bi-doped SnSe samples, the samples remain n-type over the entire temperature range of the experiments, indicating that electrons dominate the transport properties.

The temperature-dependent electrical conductivity and Seebeck coefficient were simultaneously analyzed using nonlinear least-squares minimization to a small set of effective parameters that reveal the transport properties of the system. The fitting was carried out in mathematica using a Nelder-Mead simplex algorithm, with the Seebeck coefficient and electrical conductivity assigned equal weight in the minimization procedure. The extracted parameters include the concentrations of ionized acceptors (for p-type samples) or donors (for n-type samples), and dimensionless scattering parameters describing the energy dependence of the carrier relaxation time. Additional details of the model are provided in the Supplemental Information, Sec. SI 3. The extracted ionized acceptor and donor concentrations fall within the range reported in the literature (see Table II) [11]. The fitted parameters A_n and A_p provide insight into the dominant scattering behavior for electrons and holes, respectively. Within conventional transport theory, values of $|A| \sim 2$ are commonly associated with acoustic phonon-limited scattering, while ionized-impurity scattering yields typically $|A| \sim 4$ [32].

For both pristine SnSe and the 2% Bi-doped SnSe samples, the extracted hole scattering parameters are close to 4, consistent with dominant ionized impurity scattering. In pristine SnSe, Sn vacancies acting as ionized acceptors do not appear to strongly affect electron scattering within the present analysis. In contrast, for the 2% Bi-doped SnSe sample, the fitting yields a large negative value of the electron transport parameter, $A_n \approx -10.3$. This result arises from the constraints imposed by the transport model, in which both the carrier mobilities are treated as fixed parameters. Within this framework, suppression of the electron contribution to transport cannot occur through variations in mobility. Instead, the reduction of electron transport is achieved through a shift of the effective electron chemical potential, which in the model is controlled by the parameter A_n. Consequently, a strongly negative value of A_n is required to reproduce

TABLE II. The estimated transport parameters of $Sn_{1-x}Bi_xSe$ samples are based on the two-carrier model.

	N_A^- (cm^{-3})	N_D^+(cm^{-3})	$\lvert A_n \rvert$	A_p	E_g (eV)	μ_e (cm^2 V^{-1}s^{-1})	μ_h(cm^2 V^{-1}s^{-1})
SnSe	1.05×10^{16}	NA	NA	3.86	*Pnma*: 0.84 Cmcm: 0.19	80 [33] (fixed)	120 [34] (fixed)
SnSe-2% Bi	NA	1.09×10^{16}	10.27	4.76	*Pnma*: 0.99 Cmcm: 0.27	80 (fixed)	120 (fixed)
SnSe-6% Bi (SnSeBi6%-only electrons)	NA	3.41×10^{17}	3.06	NA	Single phase: 0.23	132.87	NA

the experimentally observed Seebeck coefficient and electrical conductivity. The magnitude of A_n should therefore be interpreted as an effective fitting parameter rather than a direct representation of a specific scattering mechanism or intrinsic electronic-structure property. In this sense, the model provides a self-consistent semiempirical description of the temperature-dependent transport behavior. Similar effective parameters are commonly introduced in simplified thermoelectric transport models to capture the combined influence of multiple scattering mechanisms and band-structure effects that are not explicitly included in analytical expressions. The extracted band-gap values for the *Pnma* and *Cmcm* phases listed in Table II are in close agreement with the experimentally measured values of 0.86 eV [35] and 0.46 eV [36], respectively. Notably, the fitted *Cmcm* phase band gap critically governs the thermally activated carrier population and thus strongly influences the high-temperature transport response.

For the 6% Bi-doped SnSe sample, both a multicarrier transport model and a simplified single-carrier (electron-only) transport model were examined. The best agreement with the experimental data was obtained using the electron-only model, which does not explicitly incorporate the structural phase transition. This indicates that the dominant transport response is governed by degenerate electron conduction and can be captured without explicitly resolving the temperature-driven structural evolution. Within this framework, the extracted parameters include the concentration of ionized donors, a dimensionless scattering parameter, the effective band gap, and the electron mobility. The analysis yields a single effective band gap of approximately 0.23 eV and an electron mobility of 132 cm^2 V^{-1} s^{-1}. The reduced band gap in the 6% Bi-doped samples is also predicted by the DFT calculations presented below.

To further describe the origin of the extracted transport behavior and polarity switching, we turn to first-principles electronic structure calculations. While the two-carrier transport model captures the essential phenomenology of bipolar conduction and carrier compensation, it does not explicitly reveal how Bi substitution and temperature-driven structural evolution modify the underlying band structure. DFT calculations, therefore, provide a complementary perspective by directly resolving the evolution of the band edges, Fermi level position, and effective masses across the *Pnma* $\rightarrow$ *Cmcm* structural transition. In the following, we demonstrate that the experimentally observed transport trends and carrier-type reversal emerge from the temperature-dependent electronic structure of Bi-doped SnSe.

The experimental thermoelectric behavior of Bi-doped SnSe can be rationalized by the calculated electronic band structures. In these diagrams, the position of the Fermi level (E_F) determines the dominant charge carrier type: *n*-type (electron-dominated) behavior arises when E_F lies near the conduction-band minimum (CBM), while *p*-type (hole-dominated) conduction occurs when E_F approaches the valence-band maximum (VBM). As shown in Fig. 5, for the low-doping case ($x = 0.028$), the experimentally observed sign reversal near ~550 K precedes completion of the structural transition and is consistent with intrinsic band-gap narrowing and bipolar excitation. In the low-temperature *Pnma* phase, E_F resides at the bottom of the CBM and intersects Bi-derived conduction state at multiple points across the Brillouin zone, consistent with the observed negative Seebeck coefficient. Upon heating, the electronic structure of SnSe evolves significantly: the band gap progressively narrows and approaches the high-temperature *Cmcm* limit [Fig. 5(b)], rendering the system increasingly susceptible to intrinsic bipolar excitation. As the gap narrows substantially, the Fermi level shifts downward toward the valence band, consistent with the calculated band structure of the *Cmcm* phase. This shift is accompanied by a rapid increase in thermally generated holes, which have a lower effective mass in the emerging *Cmcm*-like electronic environment, as reflected in the steeper valence-band curvature. These holes with smaller effective mass progressively compensate for the Bi-donated electrons, and because their mobility is significantly higher, they dominate charge transport at elevated temperatures, leading to an apparent *p*-type response. This mechanism naturally explains the experimentally observed *n*-to-*p*-type reversal in the Seebeck coefficient. In stark contrast, this carrier type transition is suppressed at higher Bi concentrations ($x = 0.056$). The increased number of Bi dopants donates sufficient electrons to push E_F well into the conduction band, yielding strong degenerate *n*-type behavior. In this regime, the Fermi level lies above the CBM in both the *Pnma* and *Cmcm* phases, indicating that the structural transformation alone cannot account for the heavy electron doping. Consequently, the material retains a robust *n*-type character ($S < 0$) throughout the entire temperature range. We also examined the possibility of Bi intercalation between layers. In this model, for the sample with 5.56% Bi concentration, one Bi atom was placed in the Sn lattice, while the second was placed in the van der Waals gap. The corresponding electronic structure (see Fig. S5 in the Supplemental Material [16]) shows that Bi intercalation introduces Bi-derived states within the fundamental gap and shifts the Fermi level more deeply into the conduction band,

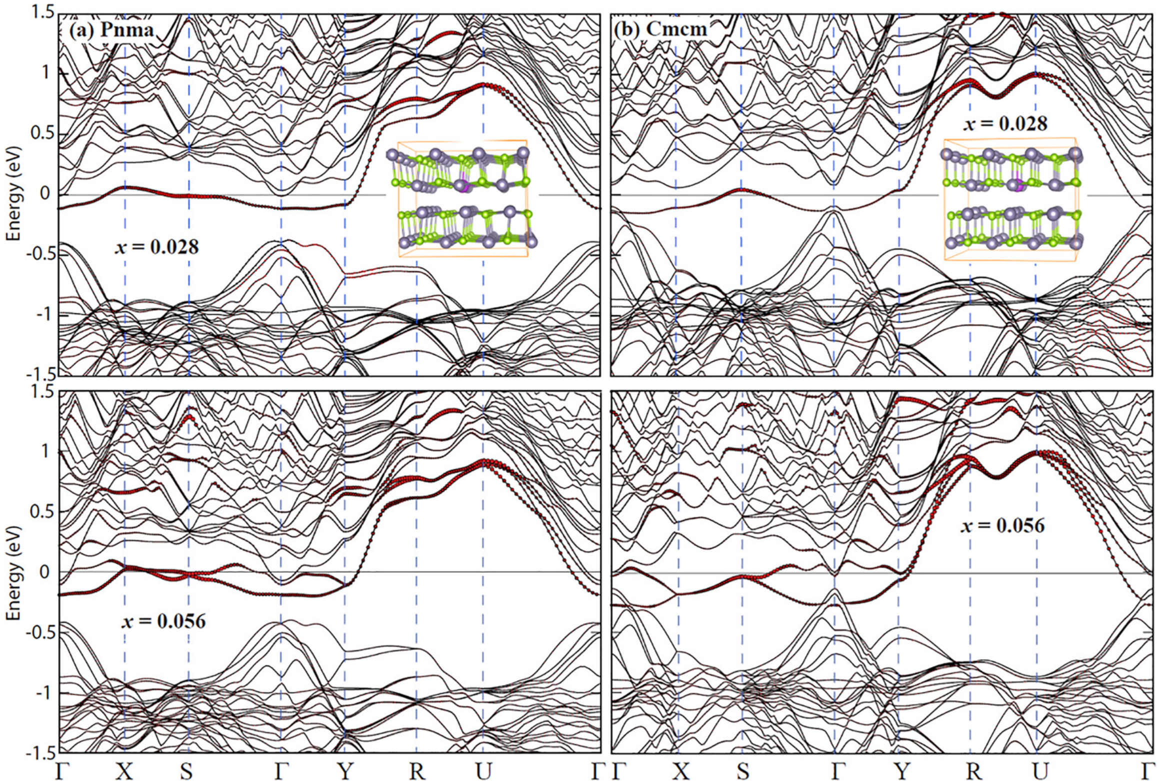

FIG. 5. Band structures of Bi-doped SnSe in (a) *Pnma* and (b) *Cmcm* phases. For each phase, the top and bottom panels correspond to Bi concentrations of $x = 0.028$ and $x = 0.056$, respectively. Red bands indicate states with Bi character. The Fermi level is set to 0 eV.

indicating strongly degenerate *n*-type behavior. This suggests that even a small degree of Bi intercalation could significantly modify both the structural signatures and electronic transport character of Bi-doped SnSe.

First-principles electron-phonon self-energy calculations further show that the fundamental band gap of the *Pnma* phase decreases monotonically with temperature, narrowing by approximately 0.1–0.15 eV between 0 and 500 K (see Fig. S6 in the Supplemental Material [16]). This gradual reduction in the gap enhances intrinsic carrier activation well before the completion of the crystallographic *Pnma* → *Cmcm* transition, providing a natural electronic origin for the onset of bipolar transport and the observed *n*- to *p*-type polarity reversal.

Taken together, the transport modeling and first-principles calculations provide a unified picture of polarity switching in Bi-doped SnSe, driven by the coupled evolution of the band structure, carrier statistics, and the structural phase transformation.

V. CONCLUSION

In conclusion, we investigated the temperature-dependent transport properties of pristine and Bi-doped SnSe across the *Pnma* → *Cmcm* structural evolution. Pristine SnSe remains *p*-type over the full temperature range, while Bi substitution progressively stabilizes *n*-type transport. At low Bi concentration (2%), the Seebeck coefficient exhibits an *n*-to-*p* polarity reversal near ∼550 K, accompanied by the emergence of bipolar transport, whereas the 6% Bi-doped sample remains *n*-type up to 880 K. A self-consistent two-carrier transport analysis captures the essential features of the nonlinear Seebeck response and conductivity evolution, highlighting the competing contributions of electrons and holes and their distinct temperature dependences.

First-principles calculations provide a definitive interpretation of these trends by directly resolving the doping- and phase-dependent electronic structure. For low Bi content, the evolution of the band edges and a downward shift of the Fermi level toward the valence band, together with progressive bandgap narrowing, naturally promote thermally activated holes that eventually dominate transport at elevated temperatures. In contrast, higher Bi concentration pushes the Fermi level deep into the conduction band, suppressing carrier compensation and yielding robust degenerate *n*-type behavior. Electron-phonon self-energy calculations further support a monotonic reduction in the *Pnma* band gap with increasing temperature, providing an electronic origin for the onset of bipolar transport prior to the completion of the crystallographic transition. Together, our combined experimental, modeling, and first-principles results establish a unified mechanism for polarity switching in Bi-doped SnSe and provide a framework for engineering carrier compensation and bipolar transport in SnSe-based systems.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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